Generative artificial intelligence enhances creativity but reduces the diversity of novel content

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Abstract. Creativity is core to being human. Generative artificial intelligence (GenAI) holds promise for humans to be more creative by offering new ideas, or less creative by anchoring on GenAI ideas. We study the causal impact of GenAI on the production of a creative output in an online experimental study where some writers are could obtain ideas for a story from a GenAI platform. Access to GenAI ideas causes an increase in the writer's creativity with stories being evaluated as better written and more enjoyable, especially among less creative writers. However, GenAI-enabled stories are more similar to each other than stories by humans alone. Our results have implications for researchers, policy-makers and practitioners interested in bolstering creativity, but point to potential downstream consequences from over-reliance.

One sentence summary: Generative artificial intelligence helps produce more creative stories—but they are also more similar to each other.

Keywords: creativity, generative artificial intelligence, machine behavior, ethics, experiment.

Creativity is fundamental to innovation and human expression through literature, art, and music (*1*). However, the emergence of generative artificial intelligence (GenAI) technologies is challenging several long-standing assumptions about the uniqueness and superiority of human-generated content (*2*). GenAI is able to create new content in text (e.g., ChatGPT), images (e.g., Midjourney), audio (e.g., Jukebox), and video (e.g., Pictory). While GenAI has previously been shown to enable joint AI-human storyline development (*3*), increase quality and efficiency of production of typical white-collar work (*4*), promote productivity in customer support relations (*5,6*), speed up programming tasks (*7*), and enhance persuasion messaging (*8*), little is known about GenAI's potential impact on a fundamental human behavior: the ability of humans to be creative.

In this paper, we study how GenAI affects participants' ability to produce creative content. The domain we focus on is literature, an "open-creativity" task. Open-creativity tasks are critical in much of human activity and the economy, but have been found to be difficult to change or improve using economic incentives (*9*). Rather than employing incentives, our paper focuses on GenAI as a potential influence that affects humans' ability to be creative on an open-creativity task.

There are at least two ways in which the availability of GenAI can affect creativity. On the one hand, GenAI may enhance: generated ideas from AI may be used as a "springboard" for the human mind, providing potential starting points that can result in a "tree structure" of different storylines (*3,10*). It can also offer multiple starting venues that help a human writer overcome "writer's block" and the fear of a blank page (*11*). If this is the case, we would expect GenAI to lead to more creative output generated by human writers.

Conversely, GenAI may hamper: by anchoring the writer to a specific idea, or starting point for a story, GenAI may restrict the variability of a writer's own ideas from the start, inhibiting open creativity. Moreover, the output offered by GenAI may be derivative and thus not provide a fertile ground for new and creative ideas. If this is the case, we would expect GenAI to lead to more similar stories and potentially less creative output generated by human writers. Note that these two pathways in which GenAI can affect human creativity may not be mutually exclusive: it is possible that GenAI enhances human's ability to be creative in some ways (e.g. novelty) but not in others (e.g. usefulness) (*12*).

This paper aims to answer these questions through a pre-registered, two-phase experimental online study (see Figure 1 for the experimental design and Methods below for details). In the first phase of our study, we recruited a group of *N*=293 participants ("writers") who are asked to write a short, eight sentence story. Participants are randomly assigned to one of three conditions: *Human only*, *Human with 1 GenAI idea*, and *Human with 5 GenAI ideas* (see Table S1 for balance across conditions). In our *Human only* baseline condition, writers are assigned the task with no mention of or access to GenAI. In the two GenAI conditions, we provide writers with the option to call upon a GenAI technology (OpenAI's GPT-4 model) to provide a three-sentence starting idea to inspire their own story writing. In one of the two GenAI conditions (*Human with 5 GenAI ideas*), writers can choose to receive up to five GenAI ideas, each providing a possibly different inspiration for their story. After completing their story, writers are asked to self-evaluate their story on novelty,

usefulness, and several emotional characteristics (see Supplementary Information (SI) Section 1 for all study questions).

In the second phase, the stories composed by the writers are then evaluated by a separate group of *N*=600 participants ("evaluators") (see Table S2 for balance across conditions). Evaluators read six randomly selected stories without being informed about writers being randomly assigned to access GenAI in some conditions (or not). All stories are evaluated by multiple evaluators on novelty, usefulness, and several emotional characteristics. After disclosing to evaluators whether GenAI was used during the creative process (*13*), we ask evaluators to rate the extent to which ownership and hypothetical profits should be split between the writer and the AI (*14*). Finally, we elicit evaluators' general views on the extent to which they believe that the use of AI in producing creative output is ethical, how story ownership and hypothetical profits should be shared between AI creators and human creators, and how AI should be credited in the involvement of the creative output (*15,16*).

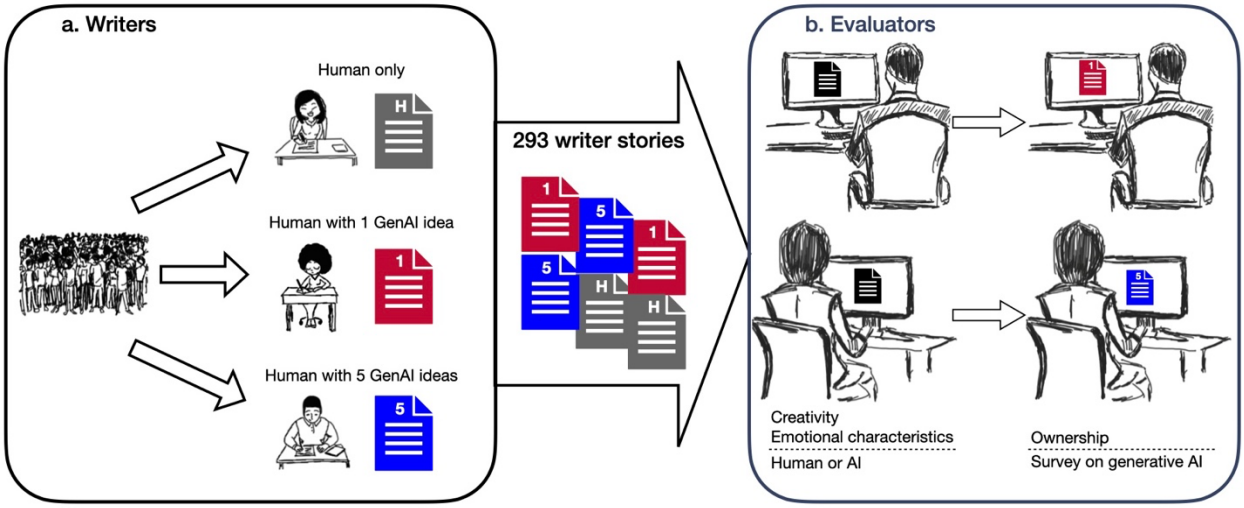

Figure 1. Visual representation of experimental design. **a**, Participants are recruited, provide consent to participate in the study, and complete the divergent association task (DAT)—a measure of an individual's inherent creativity (see Olson et al. (*17*))—before being randomly assigned to one of three experimental conditions: a *Human only* condition where the story was written with no GenAI assistance, a *Human with 1 GenAI idea* condition, and a *Human with 5 GenAI ideas* condition. A total of 293 stories are collected and then passed to evaluators. **b**, Evaluators provide ratings on six randomly assigned stories. The evaluators cycle through each story three times. First, prior to any information revelation, the evaluator assesses the creativity and emotional characteristics of the story. Second, the evaluator is asked to assess how likely the story was written by an AI versus a human. Third, the evaluator is told about whether the writer had access to and used GenAI and then provides responses about the ownership claim of the writer of each story. Evaluators then provide general responses to their views of GenAI.